

2023

reezocar

Reezocar Rethinks Car Buying Using Computer Vision and ML on AWS

Learn how automotive company Reezocar is estimating vehicle repairs swiftly and accurately using AWS HPC and ML infrastructure powered by NVIDIA.

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90 ms or less

to estimate repair costs

Dete

using c



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Overview

Every car that finds its way to a landfill marks another dent in the fight for a sustainable future. [Reezocar](#), an online hub for buying and selling used cars, has a mission to change this.

Based in France, Reezocar offers a safe and streamlined process for purchasing refurbished vehicles and helps sellers fetch fair value for their cars. By using cutting-edge machine learning (ML) solutions, it provides buyers with detailed insights into a vehicle's condition.

The technologies that underpin Reezocar's efforts are part of Amazon Web Services (AWS). Using high performance computing (HPC) and machine learning (ML) infrastructure—powered by NVIDIA GPUs—on AWS, the company can meticulously detect car dents and imperfections. Its technology can estimate repair costs in milliseconds and helps extend the serviceable life of vehicles, steering them away from premature disposal.



Opportunity | Using AWS ML Services to Power Intensive Computer Vision Workloads for Reezocar

Reezocar has one of the largest catalogs of used and new cars in Europe, with millions of cars listed on its website at any given moment. Anyone in the world can visit Reezocar to look for their preferred car and have it delivered directly to their door. Users can also list their vehicles for sale on the website or request a trade-in from Reezocar.

“We want to make sure that existing cars on the market have the longest usable life possible,” said [redacted], chief technology officer at Reezocar. “We take these vehicles, estimate their repair needs, and [redacted] most of [redacted]”



refurbishing. After they are refurbished, we decide whether to sell direct to buyers or sell them for parts to professionals.”

Using ML and computer vision, Reezocar automatically calculates repair costs by analyzing images of each vehicle. The system meticulously scans photos for damages such as dents, scratches, and paint wear. Once identified, these imperfections are cross-referenced with a vast database containing average repair costs for similar damages on comparable vehicle models. Then, the system determines the estimated cost of repairing the vehicle.

As Reezocar grew, its initial cloud provider couldn’t handle the growing demand. The infrastructure was lackluster in performance, and Reezocar wasn’t receiving adequate technical support. The company needed a robust, responsive HPC infrastructure that could use ML to process these compute-intensive workloads and produce swift and accurate estimations.

Reezocar knew that it was time for a change, so it chose to migrate to AWS. “The customer obsession at AWS is real,” says Fabre. “We have never felt like we’re alone at all when we work with the AWS team. The level of professional commitment is unmatched.”

Solution | Adopting Amazon EC2 P4d Instances to Estimate Repair Costs in 90 ms or Less

Reezocar began to explore AWS solutions by conducting a proof of concept. This initial step involved adopting [Amazon Elastic Compute Cloud](#) (Amazon EC2) [P4d Instances](#), which are powered by NVIDIA A100 Tensor Core GPUs and deliver high performance for ML training and HPC applications in the cloud. Using these instances, Reezocar generated a synthetic dataset of damaged-vehicle images by modifying three-dimensional computer-aided design models of different cars. This dataset laid the groundwork for training a deep convolutional neural network, fine-tuned to determine whether the car in a given image has sustained damage.

As images are uploaded to the Reezocar website, they are stored securely on [Amazon Simple Storage Service](#) (Amazon S3), an object storage service built to retrieve any amount of data from anywhere. The images are then funneled through [Amazon SageMaker Ground Truth](#)—a service that helps label raw data—wh

annotates images to indicate any damage.



Reezocar built an ML algorithm using [Amazon SageMaker](#), a service used to build, train, and deploy ML models for any use case. Using a combination of real-world and synthetic data, the algorithm analyzes annotated images to determine the cost of refurbishment—with exceptional accuracy. “The goal of the algorithm is to predict the cost of repairs based on the features of the car, the average price of comparable refurbishment, and business-based rules,” says Tarek Ben Charrada, lead data scientist at Reezocar. “Estimating a vehicle’s repair costs usually takes around 90 milliseconds or less on AWS.”

After evaluating the promising results of the proof of concept, Reezocar made the strategic decision to migrate to AWS. In just a few months, the company fully embraced the HPC and ML capabilities of AWS, yielding transformative results. Reezocar can not only estimate repair costs with unprecedented speed but it has also bolstered its scalability and flexibility.

Furthermore, the cost-effective nature of AWS empowers Reezocar to achieve more with less. Because the company can explore new projects without worrying about costs, it has fostered an environment ripe for innovation. Reezocar can train its models faster and refine its techniques more rapidly, keeping its estimates accurate.

Outcome | Promoting a Circular Economy with Efficient Infrastructure

By accurately detecting vehicle damage and assessing refurbishment needs on AWS, Reezocar’s trade-in processes are faster and more efficient. Ultimately, this ability extends the lifespan of selected cars by up to 5 years. This helps reduce landfill waste and promotes sustainability within the automotive industry.

Looking forward, Reezocar has major plans for its technology evolution. It wants to dive deeper into ML and generative artificial intelligence and plans to harness enterprise search capabilities using [Amazon Kendra](#), which helps users find answers faster with intelligent search powered by ML.

“We need to make sure that every single step of the customer’s journey is studied and optimized,” says Fabre. “On AWS, we can make sure that we are smarter and more efficient.”

About Reezocar

Created in 2014, Reezocar aims to facilitate the purchase and financing of cars in France and across Europe. It attracts millions of monthly visitors and helps thousands of customers buy their vehicles with confidence each year.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute portfolio with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase



model to help you best match the needs of your workload.

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Amazon EC2 P4d Instances

Amazon Elastic Compute Cloud (Amazon EC2) P4d instances deliver high performance for machine learning (ML) training and high performance computing (HPC) applications in the cloud.

[Learn more »](#)

Amazon SageMaker Ground Truth

Amazon SageMaker enables you to label raw data, such as images, text files, and videos, and generate labeled synthetic data to create high-quality datasets for training machine learning (ML) models.

[Learn more »](#)

Amazon SageMaker

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Continental on AWS



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ITALY



Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

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